# Hamza Houir Alami

(+33) 6 59 15 94 81 h.houir.alami@gmail.com
linkedin.com/in/hamza-houir-alami github.com/Hamza-67
thepnlab.com

Engineering student at Arts et Métiers ParisTech with a strong foundation in mathematical optimization, operations research, and full-stack development. Founder of **ThePnLab**, a live algorithmic trading simulation platform with 37 active users. Combining rigorous analytical training from France's top engineering preparatory program with hands-on consulting and entrepreneurial experience — pursuing graduate studies in **Operations Research** to apply optimization methods at the intersection of finance, supply chain, and technology.

## Education

- **Arts et Métiers ParisTech — ENSAM** Lille, France
  *MEng in General Engineering (Diplôme d'Ingénieur) — equiv. MSc* *Sep. 2025 – Jun. 2028 (Expected)*
  - **Top 10** French Graduate Engineering School (Grandes Écoles)
  - **Relevant Coursework:** Supply Chain & Logistics, Industrial Optimization, Production Systems, Numerical Simulation (FEM, Python), Fluid Mechanics, Industrial Organization
- **Lycée Faidherbe** Lille, France
  *CPGE PSI* — Intensive preparation in Physics, Mathematics & Engineering Science* *Sep. 2023 – Jun. 2025*
  - **Top 10** engineering preparatory school in France
  - **Relevant Coursework:** Advanced Mathematics (Analysis, Algebra, Linear Algebra), Physics (Mechanics, Thermodynamics), Engineering Science, Algorithms & Python

## Work and Leadership Experience

- **Business Manager — AMJE Lille (Junior Enterprise, Arts et Métiers)** Dec. 2025 – Present
  *Student-led engineering consultancy* *Lille, France*
  *Develop leadership, project management and client relations while supervising OR consulting missions.*
  - Led routing optimization mission for a regional distributor serving **650 retail delivery points** (Hérault dept.): modeled as a **VRP**, solved with K-Means++ zone clustering + 2-opt local search, OSRM API for real-road distances — deployed on GitHub Pages
  - Managed full commercial funnel: 286 prospects contacted → 7 studies signed (2.4% conversion); identified bottleneck at qualification stage (14.7% stage yield); designed redesigned outreach strategy targeting +33% revenue growth
- **Industrial Engineering Intern — Safran, Tooling Methods Dept. (Incoming)** Jun. 2026 – Jul. 2026
  *Aerospace manufacturing* *Morocco*
  *Contribute to process engineering and tooling methods in an industrial aerospace environment.*
  - Incoming internship: process optimization, tooling design and methods engineering for aircraft component manufacturing
- **Private Tutor — Mathematics & Physics** 2024 – Present
  *Self-Employed* *Lille, France*
  *Competitive exam preparation for baccalauréat and CPGE-level students.*
  - Adapted teaching methods to individual learning profiles; focused on analytical reasoning and exam strategy

## Selected Research & Development Projects

- **ThePnLab — Algorithmic Paper Trading Platform** 2025 – Present
  *thepnlab.com | github.com/Hamza-67/ThePnLab* *Individual Project*
  *Build a full-stack platform for algorithmic trading simulation; study decision-making divergence between bot and human behavior.*
  - Daily decision cycle: composite score on **9 technical signals** selects positions across 36+ assets; **Kelly Criterion** sizes them; **VIX regime controller** scales exposure; LLM (Gemini 2.0 Flash) generates structured trade rationale
  - **Stack:** Python, FastAPI, Streamlit, PostgreSQL — **37 registered users**, live at thepnlab.com; 30-day live results: bot **+$386 PnL**, win rate 37% — profitable via favorable gain/loss ratio despite sub-50% hit rate
- **Flywheel Shape Optimization (TIPE — ENSAM Lille)** 2025
  *Mathematical Optimization | Python (SciPy, NumPy)* *Individual Research · Score 8/10*
  *Formulate and solve a constrained shape optimization problem; validate computationally and experimentally.*
  - Maximized moment of inertia $I = \mu \int r^3 \, d\theta$ under fixed-perimeter constraint; KKT conditions and the isoperimetric inequality analytically confirm the **circle as unique global maximizer**
  - Implemented and benchmarked **4 parameterizations** (Fourier, polynomial, polar, 3D surface) using **L-BFGS-B** and **Trust-Region** solvers (`scipy.optimize`); identified Runge instability above polynomial degree 20
  - Experimental validation on a 34g miniature vehicle: **1.8%–5.7% braking distance error** vs. theory ($n = 5$ runs, $p < 0.0001$)

- **Predictive Maintenance & Optimal Decision Threshold (LISPEN Lab)** Apr. 2026 – ongoing
  *Deep Learning | Python, Keras, SciPy · LISPEN Research Lab, ENSAM Lille Team · Supervised by Prof. Stephan*
  *Build a CNN fault classifier; formulate threshold selection as a formal cost-minimization problem.*
  - CNN on **mel-spectrograms** (CWRU bearing dataset): classifies fault types from raw vibration signals; preprocessing pipeline handles real sensor variability
  - Decision layer: find $\theta^*$ minimizing $\mathbb{E}[\text{cost}] = c_{\text{prev}} \cdot P(\text{FA}) + c_{\text{fail}} \cdot P(\text{MD})$ subject to machine availability $\geq 95\%$

## Technical & Language Skills

**Programming:** Python (NumPy, SciPy, Pandas, Keras), FastAPI, Streamlit, PostgreSQL, Git

**Software:** CATIA / 3DEXPERIENCE, Star-CCM+, Abaqus, LaTeX

**French:** Native

**Arabic:** Native

**English:** Fluent — TOEIC (May 2026)

**Spanish:** Intermediate (B1)

## Hobbies & Interests

**Football:** Competed at international youth level — **Ibercup Cascais** (Portugal), 40+ nations, 100+ teams; coached an amateur team during injury; developing tactical and data-driven analysis of the game.

**Geopolitics:** Regular reading and self-directed research on international affairs, regional dynamics (Middle East, North Africa, European politics) and global power shifts — building an analytical lens beyond the engineering curriculum.